

Notes: Nanda, Wang, and Zheng (RFS, 2004)

Family Values and the Star Phenomenon: Strategies of Mutual Fund Families

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1 Big picture

- **Question.** How does mutual fund family membership affect investor flows, family strategy, and subsequent fund performance?
- **Core fact.** Mutual funds are usually not stand-alone products. They are sold inside families that share reputation, advertising, distribution, service, and investor search costs.
- **Main hypothesis.** A star fund can generate positive spillovers within its family: investors attracted by one outstanding fund may also put money into other funds in the same family.
- **Why this matters.** If family-level spillovers are important, then the payoff to creating one star fund is larger than the payoff to a stand-alone fund. This changes family-level incentives.
- **Strategic implication.** Families with lower investment ability may pursue high-variance, star-creating strategies because the convex flow-performance relation rewards extreme winners more than it punishes losers.
- **Empirical setting.** The paper uses CRSP open-end mutual fund data from 1992 to 1998, focusing on diversified U.S. equity funds with family identity information.
- **Main spillover result.** A star fund increases inflows not only into the star fund itself but also into other funds in the family.
- **Magnitude result.** Compared with a stand-alone star fund, a star fund in an average-size family with about seven funds can generate a family-level inflow effect more than three times as large.
- **Star-creating strategy result.** Families with higher cross-fund return dispersion and a larger number of funds are more likely to produce star funds.
- **Ability result.** Not all star families have superior future performance. Families that generate stars despite a low ex ante probability of doing so appear to be the true high-ability families.
- **Investor warning.** Naively chasing star families is not a reliable strategy. A more discerning strategy is to identify stars that are unlikely to have arisen from a high-variance family strategy.

2 Incremental contribution of the paper

2.1 Contribution relative to the flow-performance literature

The mutual fund flow literature had already documented a convex relation between individual fund performance and investor flows: very strong performance attracts large inflows, while poor performance does not create symmetrically large outflows. This paper adds a family-level dimension to that finding.

- Earlier work mostly treated each fund as an independent product.
- Nanda, Wang, and Zheng show that the performance of one fund affects flows to other funds managed by the same family.
- This implies that the unit of analysis should sometimes be the fund family, not just the individual fund.
- The paper therefore reframes the flow-performance relation as a family-wide incentive problem.

Interpretation: The incremental idea is that convex flow-performance incentives are amplified by family spillovers. This amplification changes the strategic incentives of the fund complex.

2.2 Contribution relative to mutual fund family research

Prior research recognized that fund families provide distribution economies and brand value. This paper shows that the family structure also creates within-family externalities.

- A star fund can act like an advertisement for the entire family.
- Investors who learn about the family through a star fund may invest in nonstar sibling funds.
- Family reputation and investor search costs make this spillover plausible.
- The paper quantifies this effect using fund-level and family-level panel regressions.

Interpretation: The family is not just an administrative umbrella. It is an economic unit that shapes strategy and cash flows.

2.3 Contribution relative to the literature on mutual fund skill

The paper also contributes to the debate over whether star performance reflects skill.

- A star can arise from genuine skill.
- A star can also arise from a high-variance strategy or from managing many funds, which increases the chance that one fund lands in the top tail.
- The paper separates these possibilities by measuring the ex ante probability of producing stars.
- Families with low cross-fund return dispersion and a small number of funds have lower ex ante star probabilities.
- When such families produce stars, the star is more likely to indicate true investment ability.

Interpretation: The incremental contribution is a sorting logic: not all stars are equal. The same observed star performance has different informational content depending on the family strategy that generated it.

2.4 Contribution to managerial strategy and incentives

The paper connects investor-flow incentives to deliberate family strategy.

- Families can increase the probability of a star by raising dispersion in investment strategies across their member funds.
- This strategy creates a lottery-like payoff: one winner can produce large inflows for the whole family.
- Lower-ability families may have stronger incentives to adopt this strategy because they have less to gain from skill-based investing.
- High-ability families may prefer lower-variance strategies that exploit their comparative investment advantage.

Interpretation: The paper turns the star phenomenon into a strategic choice problem. Families can design their product line and portfolio dispersion to increase the chance of producing marketable winners.

2.5 Incremental empirical contribution

The paper's empirical contribution is broad rather than relying on a single test.

- It documents family-level spillovers from stars to aggregate family flows.
- It documents spillovers from stars to nonstar funds inside the same family.

- It shows that cross-fund return dispersion and family size predict the probability of producing a star.
- It verifies robustness using Morningstar-style five-star ratings.
- It studies subsequent portfolio performance to distinguish skill from star-creating strategy.

Interpretation: The paper builds a full chain: star performance attracts flows, family strategy affects the probability of creating stars, and only some types of stars forecast future performance.

3 Data and measurement

3.1 Sample construction

- The data come from the CRSP mutual fund database.
- The sample covers U.S. open-end diversified equity funds from January 1992 to December 1998.
- The paper requires family identity, monthly total net assets, monthly returns, and annual fund characteristics.
- The sample excludes sector funds, international funds, and balanced funds.
- Multiple share classes are adjusted so that the same underlying fund is not counted multiple times when identifying stars and dogs.

Interpretation: The focus on diversified U.S. equity funds makes the Fama–French risk-adjustment model more appropriate and keeps the family strategy comparison relatively clean.

3.2 Risk-adjusted performance

For each fund i , the paper estimates Fama–French three-factor adjusted performance using:

$$R_{it} - RF_t = \alpha_i + \beta_i^{RMRF} RMRF_t + \beta_i^{SMB} SMB_t + \beta_i^{HML} HML_t + e_{it}. \quad (1)$$

The monthly three-factor adjusted return is:

$$a_{it} = \alpha_i + e_{it}. \quad (2)$$

Family-level adjusted performance is the total-net-assets-weighted average across member funds:

$$a_{ft} = \frac{\sum_{i=1}^n a_{it} TNA_{it}}{\sum_{i=1}^n TNA_{it}}. \quad (3)$$

Interpretation: The paper uses risk-adjusted returns to avoid classifying funds as stars merely because they take standard factor exposures.

3.3 New money and new money growth

For each fund, new money is the change in total net assets net of asset appreciation:

$$Newmoney_{it} = TNA_{it} - TNA_{i,t-1}(1 + R_{it}). \quad (4)$$

Fund-level new money growth is:

$$Newmoneygrowth_{it} = \frac{Newmoney_{it}}{TNA_{i,t-1}}. \quad (5)$$

Family-level new money is the sum across all member funds:

$$Newmoney_{ft} = \sum_{i=1}^n Newmoney_{it}. \quad (6)$$

Family-level new money growth is:

$$Newmoneygrowth_{ft} = \frac{Newmoney_{ft}}{\sum_{i=1}^n TNA_{i,t-1}}. \quad (7)$$

Interpretation: This measure isolates investor cash flows from performance-driven asset growth. It is the key dependent variable for testing whether stars attract new money.

3.4 Family strategy: cross-fund return dispersion

The paper proxies family investment strategy using the cross-fund standard deviation of adjusted returns:

$$CrossFundStd_{ft} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (a_{it} - a_{ft})^2}. \quad (8)$$

Interpretation: A high cross-fund standard deviation means that funds in the same family follow more dispersed strategies. This raises the chance that at least one fund lands in the extreme right tail.

3.5 Star and dog definitions

- For each month, the paper computes each fund’s average risk-adjusted return over the previous 12 months.
- The top 5% are defined as star funds.
- The bottom 5% are defined as dog funds.
- A star family is a family with at least one star fund.
- A dog family is a family with at least one dog fund.

Interpretation: The top-tail and bottom-tail definitions allow the paper to examine both positive spillovers from stars and possible negative effects from poor performers.

3.6 Morningstar-based stars

The paper also replicates Morningstar five-star ratings.

- Morningstar ratings are well-known to investors and often used in fund marketing.
- Only funds with sufficient performance history receive ratings.
- The authors replicate Morningstar’s rating procedure using return and risk components.
- They compare their May 1995 replication with Morningstar Principia Pro Plus and find close agreement.

Interpretation: Morningstar stars provide an alternative, industry-relevant star definition. Similar results under this definition strengthen the interpretation that investors respond to salient star signals.

4 Empirical specifications and methodology

4.1 Family-level flow regression

The first main test examines whether a star increases family-level new money growth:

$$Newmoneygrowth_{f,t} = \alpha_f + \beta_1 PastPerformance_{f,t-12:t-1} + \beta_2 CrossFundStd_{f,t-1} \quad (9)$$

$$+ \beta_3 NumberOfFunds_{f,t-1} + \beta_4 FamilySize_{f,t-1} + \beta_5 Turnover_{f,t-1} \quad (10)$$

$$+ \beta_6 ExpenseRatio_{f,t-1} + \beta_7 FrontEndLoad_{f,t-1} \quad (11)$$

$$+ \beta_8 StarFamilyDummy_{f,t-12:t-1} + \beta_9 MSStarFamilyDummy_{f,t-12:t-1} \quad (12)$$

$$+ \beta_{10} DogFamilyDummy_{f,t-12:t-1} + \varepsilon_{f,t}. \quad (13)$$

Interpretation: The coefficient on the star-family dummy captures whether having at least one star increases aggregate family flows beyond the effects of past family performance and other controls.

4.2 Fund-level spillover regression

The second main test examines whether star performance affects flows to other funds in the family:

$$Newmoneygrowth_{i,t} = \alpha_i + \beta_1 PastPerformance_{i,t-12:t-1} + \beta_2 FundSize_{i,t-1} \quad (14)$$

$$+ \beta_3 Turnover_{i,t-1} + \beta_4 ExpenseRatio_{i,t-1} + \beta_5 FrontEndLoad_{i,t-1} \quad (15)$$

$$+ \beta_6 StarFundDummy_{i,t-12:t-1} + \beta_7 StarFamilyDummy_{i,t-12:t-1} \quad (16)$$

$$+ \beta_8 MSStarFundDummy_{i,t-12:t-1} + \beta_9 MSStarFamilyDummy_{i,t-12:t-1} \quad (17)$$

$$+ \beta_{10} DogFundDummy_{i,t-12:t-1} + \beta_{11} DogFamilyDummy_{i,t-12:t-1} + \varepsilon_{i,t}. \quad (18)$$

Interpretation: The key spillover coefficient is the family-star dummy after controlling for whether the fund itself is a star. A positive coefficient means nonstar sibling funds receive inflows when the family has a star.

4.3 Probability of producing stars

The paper estimates logit models for the probability that a family produces at least one star:

$$S_{f,t} = \beta_0 + \beta_1 CrossFundStd_{f,t} + \beta_2 FamilySize_{f,t} + \beta_3 NumberOfFunds_{f,t} \quad (19)$$

$$+ \beta_4 Turnover_{f,t} + \beta_5 ExpenseRatio_{f,t} + \beta_6 FrontEndLoad_{f,t} \quad (20)$$

$$+ \beta_7 LaggedStarDummy_{f,t} + \beta_8 LaggedDogDummy_{f,t} + u_{f,t}. \quad (21)$$

Interpretation: The cross-fund standard deviation and number of funds are central. They test whether families can increase the chance of producing a star by increasing strategy dispersion or by operating more funds.

4.4 Portfolio formation tests

The portfolio tests sort families based on three characteristics measured in the previous 12 months:

1. cross-fund standard deviation, high versus low;
2. number of member funds, large versus small;
3. star identity, star versus nonstar.

Families are then sorted into portfolios such as:

- HighStd-Large-Star;

- HighStd-Small-Star;
- LowStd-Large-Star;
- LowStd-Small-Star;
- analogous nonstar portfolios.

Interpretation: A low-standard-deviation small family has a low ex ante probability of producing a star. If such a family produces a star, the star is more likely to reveal skill.

4.5 Performance measurement in portfolios

The paper evaluates future portfolio performance using:

- one-factor adjusted monthly returns;
- three-factor adjusted monthly returns;
- four-factor adjusted monthly returns.

Portfolios are formed with annual and monthly rebalancing.

Interpretation: The portfolio tests ask whether star performance forecasts future returns and whether the forecast depends on the family's star-producing probability.

5 Identification and empirical design

5.1 What identifies star spillovers

- The paper compares flows to star and nonstar families while controlling for past performance and family characteristics.
- Family fixed effects remove persistent differences across families.
- Fund-level regressions separate own-star effects from sibling-star spillovers.
- Dog-family indicators test whether bad outliers create negative spillovers.

Interpretation: The strongest evidence for spillovers comes from positive coefficients on star-family variables in fund-level regressions after controlling for a fund's own performance and own star status.

5.2 Why cross-fund dispersion matters

- Families with more dispersed fund strategies have a higher probability of producing extreme winners.
- The same dispersion can reflect deliberate star-creating strategy rather than superior skill.
- If high-dispersion star families subsequently underperform, this supports the strategy interpretation.

Interpretation: Cross-fund dispersion is both a strategy variable and a device for distinguishing true ability from lottery-like star creation.

5.3 Incremental interpretation of portfolio tests

- The portfolio tests do not just ask whether star families outperform.
- They ask whether stars with low ex ante probability of occurrence outperform stars with high ex ante probability of occurrence.
- This design creates a more meaningful test of skill because high-dispersion large families are mechanically more likely to produce stars.

Interpretation: The paper’s portfolio logic is close to a signal-quality test: the informational content of a star is higher when it is harder to produce mechanically.

5.4 Limitations

- Star definitions are based on past returns and can be sensitive to risk-adjustment models.
- Family strategy is proxied by cross-fund return dispersion, which may reflect risk or style differences rather than deliberate policy.
- The sample period is 1992–1998, which limits long-run generality.
- The paper does not observe marketing intensity directly.
- The analysis is observational rather than a randomized experiment.

Interpretation: The paper’s strength is the internal coherence of its evidence: flow spillovers, star-probability determinants, and future performance patterns all point to a family-level star strategy.

6 Tables and figures

6.1 Table 1 and Table 2: sample structure and star-family characteristics

Read:

- Table 1 shows that the number of fund families increases from 178 in 1992 to 278 in 1998.
- Average family size rises from 3.88 funds to 6.83 funds.
- Average family TNA rises from \$1.64 billion to \$4.12 billion.
- Table 2 shows that star funds and star families are present in every year and that many star families are high-standard-deviation families.
- The number of star families rises from 34 in 1992 to 52 in 1998.

Economic message: The fund family structure becomes larger and more important during the sample period, creating a meaningful environment for family-level spillovers.

Table 1
Annual summary statistics for mutual fund families, 1992-1998

Year	Number of families	Number of funds per family	New money growth (%)	Family TNA (\$ mil.)	Average TNA per fund (\$ mil.)	Turnover ratio	Expense ratio (%)	Front-end load (%)	Three-factor adjusted return (%)	Standard deviation of return (%)
Mean										
1992	178	3.88	1.71	1638.09	275.43	0.76	1.30	2.14	-0.05	1.08
1993	199	4.30	1.73	2017.09	310.88	0.72	1.28	1.90	0.10	1.22
1994	216	4.56	1.20	2127.50	311.14	0.77	1.33	1.66	-0.18	1.05
1995	223	5.03	1.35	2553.43	360.60	0.78	1.30	1.55	-0.16	1.21
1996	238	5.22	1.45	3395.60	453.90	0.78	1.30	1.40	-0.07	1.15
1997	253	5.81	1.92	3616.58	471.18	0.74	1.27	1.20	-0.30	1.30
1998	278	6.83	1.11	4121.75	456.25	0.85	1.29	1.21	-0.23	1.47
Standard deviation										
1992	—	3.10	3.76	4925.36	496.40	1.35	0.67	2.42	0.39	0.49
1993	—	4.02	3.36	6766.54	592.47	0.81	0.53	2.30	0.65	0.41
1994	—	4.70	4.20	6408.60	606.70	0.72	0.69	2.19	0.47	0.48
1995	—	5.02	4.02	7004.37	729.02	1.05	0.60	2.12	0.50	0.57
1996	—	5.91	5.50	9415.96	899.13	1.05	0.60	1.99	0.42	0.46
1997	—	6.74	5.63	8960.67	776.04	0.59	1.12	1.84	0.54	0.55
1998	—	7.86	4.54	8978.18	748.99	1.28	0.84	1.80	0.60	0.66

The table reports annual summary statistics for mutual fund families in the period 1992-1998. Our sample excludes all single-fund families. We adjust for multiple share classes so that we only count the same fund once. For each year in the sample period, the table shows the means and standard deviations of the following family characteristics: the year-end number of mutual fund families, the number of funds within a family, the new money growth (%), the family TNA (in million dollars), the average TNA per member fund (in million dollars), the turnover ratio, the expense ratio (%), the front-end load (%), and the cross-fund standard deviation of three-factor adjusted returns (%). The number of fund families, turnover ratio, expense ratio, and front-end load are year-end measures. All other variables are measured on a monthly basis. For each mutual fund family in the sample, we first calculate the monthly averages of these variables in any given year. Then we calculate the means and standard deviations of the monthly averages across all families.

Table 1: Annual summary statistics

Table 2
Summary statistics for star funds and star families, 1992-1998

Year	Number of funds	Number of families	Average fund-level performance (%)	Average family size	Average return std. dev. (%)	Number of high std. dev. and large families	Number of low std. dev. and large families	Number of high std. dev. and small families	Number of low std. dev. and small families	Number of families with one member fund
1992	39	34	1.17	5.56	1.34	14	4	6	2	5
1993	46	40	1.23	6.80	1.57	22	5	9	2	2
1994	53	41	1.07	6.87	1.31	12	5	10	3	5
1995	60	48	1.39	7.64	1.47	17	6	16	2	4
1996	66	53	1.27	8.50	1.53	22	2	19	4	2
1997	73	64	0.83	8.43	1.47	22	7	15	11	2
1998	72	52	1.33	10.03	1.73	25	7	12	5	0

The table presents summary statistics for the star funds and star families identified at the end of each calendar year in the sample period. Star funds are defined as the top 5% performers with the highest average three-factor adjusted returns over the previous 12 months. Star families are defined as the fund families that have at least one star fund under management. Both the number of star funds and the number of star families are year-end measures. For each star fund in any given year, we compute the monthly average of three-factor adjusted returns during the year. The average performance of star funds for that year is then measured by the mean of these monthly averages across all star funds. The average size and return standard deviation of star families are defined in a similar manner. For each star family in any given year, we first compute the monthly average of the number of member funds and the monthly average of cross-fund standard deviations of three-factor adjusted returns during the year. The average size and standard deviation of star families for that year are then measured by the means of the corresponding monthly averages across all star families. We also count the number of star families within each of the following groups: high standard deviation and large families, low standard deviation and large families, high standard deviation and small families, low standard deviation and small families, and single-fund families. For any given year, a fund family is defined as a large (small) family if the monthly average of number of funds under management is above (below) that of the median family. Similarly a fund family is defined as a high (low) standard deviation family if the monthly average of cross-fund standard deviations of three-factor adjusted returns is above (below) that of the median family.

Table 2: Star funds and star families

6.2 Table 3: star-family effect on family-level flows

Read:

- The star-family dummy is positive and significant under three-factor, one-factor, Morningstar, and combined star definitions.
- In the three-factor specification, the star-family coefficient is about 0.37, with t -statistics around 2.5.
- The Morningstar-star-family coefficient is much larger, around 0.93-0.94 in the MS-star columns.
- Dog-family effects are small and generally insignificant.
- Past performance remains a strong determinant of flows.

Economic message: A star fund increases aggregate family inflows, not just the inflow to the individual star. The family captures a spillover benefit from one member fund's success.

6.3 Table 4: star effects on fund-level flows

Read:

- A fund's own star status strongly increases its own new money growth.
- The star-family dummy is also positive and significant, meaning nonstar funds benefit from a sibling star.
- Morningstar star-family effects are also positive.
- Fund size is strongly negative, consistent with smaller funds receiving higher percentage flow growth.

- Dog-fund effects are negative, but dog-family spillovers are weak.

Economic message: The fund-level regressions provide the cleanest evidence of within-family spillovers: other funds in the same family receive new money when the family has a star.

The star-fund effect on family-level new money growth

Independent variables	Dependent variables Family new money growth (<i>t</i>)			
	Three-factor star	One-factor star	MS star	Combined
Past performance	204.35*** (11.24)	200.36*** (10.68)	159.21*** (8.84)	157.73*** (8.48)
Cross-fund std. dev.	7.23 (1.26)	7.40 (1.29)	13.40** (2.56)	13.47** (2.57)
Number of funds	-1.11*** (-4.43)	-1.10*** (-4.33)	-0.94*** (-3.83)	-0.94*** (-3.80)
Family size	-2.25*** (-12.42)	-2.26*** (-12.46)	-2.25*** (-12.40)	-2.25*** (-12.41)
Turnover ratio	-0.54** (-2.12)	-0.54** (-2.12)	-0.57** (-2.23)	-0.57** (-2.24)
Expense ratio	-25.19*** (-2.80)	-24.85*** (-2.77)	-24.06*** (-2.87)	-24.04*** (-2.87)
Front-end load	-45.14*** (-5.07)	-44.94*** (-5.04)	-45.25*** (-5.16)	-45.19*** (-5.15)
Star family dummy	0.37** (2.54)	0.37** (2.55)	0.23* (1.66)	0.23* (1.67)
MS star family dummy			0.94*** (7.00)	0.94*** (7.01)
Dog family dummy	-0.15 (-1.01)	-0.15 (-1.01)	-0.08 (-0.54)	-0.15 (-1.04)
No. of observations	15,769	15,769	15,769	15,769
Adjusted R ²	0.06	0.06	0.06	0.06

The table examines the star fund effect on the family-level new money growth. We estimate the following fixed-effect panel regressions:

$$\begin{aligned}
 (Newmoneygrowth)_{f,t} = & \alpha_1 + \beta_1(past\ performance)_{f,t-12,t-1} + \beta_2(cross-fund\ std.\ dev.)_{f,t-1} \\
 & + \beta_3(number\ of\ funds)_{f,t-1} + \beta_4(family\ size)_{f,t-1} + \beta_5(turnover\ ratio)_{f,t-1} \\
 & + \beta_6(expense\ ratio)_{f,t-1} + \beta_7(front-end\ load)_{f,t-1} \\
 & + \beta_8(star\ family\ dummy)_{f,t-12,t-1} + \beta_9(MS\ star\ family\ dummy)_{f,t} \\
 & + \beta_{10}(dog\ family\ dummy)_{f,t-12,t-1} + \epsilon_{f,t}.
 \end{aligned}$$

Here, *f* and *t* denote family and month, respectively. *Newmoneygrowth* is the new money flow into the family normalized by the previous month-end family TNA. *Past performance* is measured by the average family-level risk-adjusted return over the previous 12 months. *Cross-fund std. dev.* is defined as the standard deviation of three-factor adjusted returns across all funds in the family. *Number of funds* is the logarithm of the total number of member funds managed by the family. *Family size* is measured by the logarithm of average TNA per member fund relative to the TNA per fund of the median family. *Turnover ratio*, *expense ratio*, and *front-end load* are defined as the TNA-weighted averages of the corresponding fund-level measures. *Star family dummy* and *dog family dummy* are dummy variables indicating whether the family has at least one star or dog fund under management. *MS star family dummy* is a dummy variable indicating whether the family has at least one Morningstar five-star fund under management. The columns labeled three-factor (one-factor) star report coefficient estimates when three-factor (one-factor) adjusted returns are used to measure past performance, to calculate cross-fund standard deviation of returns, and to identify star and dog families. In the columns labeled MS star, while using three-factor adjusted returns to define past performance and standard deviation, we use Morningstar rankings instead to identify star families. The last column of the table (labeled Combined) reports coefficient estimates when both star measures (three-factor stars and MS stars) are included in the regression. As in the previous column, we use three-factor adjusted returns to measure past performance and to calculate standard deviation. The reported coefficient estimates and standard errors are adjusted for autocorrelation (AR1), contemporaneous correlation across panels, and heteroscedasticity. *T*-statistics are reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and *** respectively. All coefficients

The star fund effect on fund-level new money growth

Independent variables	Dependent variables Fund new money growth (<i>t</i>)			
	Three-factor star	One-factor star	MS star	Combined
Past performance	187.55*** (27.89)	180.84*** (24.96)	156.04*** (30.50)	149.04*** (27.50)
Fund size	-4.81*** (-82.84)	-4.82*** (-82.67)	-4.77*** (-82.68)	-4.77*** (-83.78)
Turnover ratio	-0.81*** (-7.71)	-0.81*** (-7.70)	-0.79*** (-7.56)	-0.79*** (-7.55)
Expense ratio	-44.58*** (-5.01)	-44.56*** (-5.01)	-46.06*** (-5.23)	-46.82*** (-5.32)
Front-end load	-13.44*** (-2.28)	-13.42*** (-2.28)	-13.59*** (-2.31)	-12.48*** (-2.33)
Star fund dummy	1.04*** (5.44)	1.07*** (5.58)	1.66*** (9.35)	1.66*** (9.50)
Star family dummy	0.37*** (4.08)	0.37*** (4.09)	0.19** (2.26)	0.19** (2.25)
MS star fund dummy			3.09*** (19.96)	3.09*** (19.95)
MS star family dummy			0.22** (2.36)	0.22** (2.35)
Dog fund dummy	-0.56*** (-2.96)	-0.81*** (-4.32)	-0.51*** (-2.73)	-0.55*** (-2.94)
Dog family dummy	0.02 (0.23)	-0.03 (-0.32)	0.01 (0.08)	0.02 (0.24)
No. of observations	141663	141663	141663	141663
Adjusted R ²	0.11	0.11	0.11	0.11

The table examines the spillover effect associated with the star performance. We estimate the following fixed-effect panel regressions:

$$\begin{aligned}
 (Newmoneygrowth)_{i,t} = & \alpha_i + \beta_1(past\ performance)_{i,t-12,t-1} + \beta_2(fund\ size)_{i,t-1} + \beta_3(turnover\ ratio)_{i,t-1} \\
 & + \beta_4(expense\ ratio)_{i,t-1} + \beta_5(front-end\ load)_{i,t-1} + \beta_6(star\ fund\ dummy)_{i,t-12,t-1} \\
 & + \beta_7(star\ family\ dummy)_{i,t-12,t-1} + \beta_8(MS\ star\ fund\ dummy)_{i,t} \\
 & + \beta_9(MS\ star\ family\ dummy)_{i,t} + \beta_{10}(dog\ fund\ dummy)_{i,t-12,t-1} \\
 & + \beta_{11}(dog\ family\ dummy)_{i,t-12,t-1} + \epsilon_{i,t}.
 \end{aligned}$$

Here, *i* is the index for individual fund and *t* is the index for month. *Past performance* is measured by the average fund-level risk-adjusted return over the previous 12 months. *Fund size* is defined as the logarithm of fund-level TNA relative to the TNA of the median fund. Dummy variables and *front-end load* are all fund-level statistics. The remaining six variables are *dummies*. *Star fund dummy* and *MS star fund dummy* equal one if the fund itself is a star or MS star fund, while *star family dummy* and *MS star family dummy* equal one if the fund is not a star or MS star but belongs to a star or MS star family. Similar definitions apply to *dog fund dummy* and *dog family dummy*. The columns labeled three-factor (one-factor) star report coefficient estimates when three-factor (one-factor) adjusted returns are used to measure past performance and to identify stars and dogs. In the columns labeled MS star, while using three-factor adjusted returns to define past performance and dogs, we use Morningstar rankings instead to identify star funds and star families. The last column of the table (labeled Combined) reports coefficient estimates when both star measures (three-factor stars and MS stars) are included in the regression. As in the previous column, we use three-factor adjusted returns to measure past performance and to identify dogs. The reported coefficient estimates and standard errors are adjusted for autocorrelation (AR1) and heteroscedasticity in the residuals. *T*-statistics are reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and *** respectively. All coefficients reported in the table are the actual coefficient estimates multiplied by 100.

6.4 Table 5 and Table 6: probability of producing stars

Read:

- Table 5 shows that cross-fund standard deviation significantly increases the probability of producing top-5% three-factor stars in most years.
- The number of funds also raises star probability in several years.
- Family size has a weaker and less consistent relation.
- Table 6 repeats the analysis for Morningstar-style stars.
- For Morningstar stars, cross-fund standard deviation is less consistently important, while the number of funds is often positive and significant.

Economic message: Families can increase the odds of creating stars either by operating more funds or by increasing dispersion across strategies. This supports the idea that star creation can be partly strategic.

	1992	1993	1994	1995	1996	1997	1998
Constant	-5.61*** (-4.77)	-5.70** (-4.07)	-3.59*** (-3.39)	-5.28*** (-5.06)	-4.28*** (-4.76)	-3.13*** (-4.40)	-3.96*** (-5.06)
Cross-fund std. dev.	100.60** (2.18)	138.00** (2.97)	135.70*** (2.66)	180.10*** (3.34)	223.60*** (4.14)	111.80** (2.45)	49.61 (1.25)
Family size	0.49** (2.23)	0.42* (1.89)	0.13 (0.64)	0.14 (0.78)	0.03 (0.22)	-0.17 (-1.22)	0.04 (0.28)
Number of funds	0.98** (2.05)	1.57*** (3.48)	0.35 (0.86)	1.26*** (3.36)	0.52 (1.58)	0.53* (1.82)	0.38 (1.40)
Turnover ratio	0.08 (0.58)	0.88* (1.90)	0.68 (1.59)	-0.42 (-0.96)	-0.01 (-0.04)	-0.40 (-1.10)	0.29 (1.10)
Expense ratio	68.97* (1.95)	-63.01 (-0.80)	-79.42 (-1.29)	-10.34 (-0.20)	-41.80 (-0.93)	-13.99 (-0.82)	40.54 (1.52)
Front-end load	4.16 (0.43)	-13.50 (-1.14)	0.71 (0.07)	-8.41 (-0.78)	-21.63* (-1.85)	-11.93 (-1.17)	9.28 (0.90)
Lagged star dummy	0.61 (1.04)	1.25** (2.08)	1.18** (2.18)	0.30 (0.57)	0.57 (1.25)	0.13 (0.31)	1.22** (3.08)
Lagged dog dummy	0.46 (0.68)	-0.72 (-0.67)	-0.14 (-0.21)	-0.80 (-1.16)	-0.32 (-0.57)	0.67 (1.45)	-0.80 (-1.46)

The table examines which factors contribute to increasing the odds of producing stars. For each year from 1992 to 1998, we apply the following logit model to examine how family characteristics are related to the probability of producing star funds:

$$S_j = \beta_0 + \beta_1(\text{cross-fund std. dev.})_j + \beta_2(\text{family size})_j + \beta_3(\text{number of funds})_j + \beta_4(\text{turnover ratio})_j + \beta_5(\text{expense ratio})_j + \beta_6(\text{front-end load})_j + \beta_7(\text{lagged star dummy})_j + \beta_8(\text{lagged dog dummy})_j.$$

Here, f is the index for fund family and S is an indicator variable that equals one if the family has at least one star fund and zero otherwise. *Cross-fund std. dev.* is the monthly average of the cross-fund standard deviations of three-factor adjusted returns for the family. *Family size* is the logarithm of monthly average of the relative family size (measured by the average TNA per member fund relative to that of the median family). *Number of funds* is the logarithm of monthly average of the total number of member funds managed by the family. *Turnover ratio*, *expense ratio*, and *front-end load* are the monthly averages of the corresponding statistics for the family. *Lagged star dummy* is a dummy variable that equals one if the family has at least one star fund in the previous year. *Lagged dog dummy* is a dummy variable that equals one if the family has at least one dog fund in the previous year. In this regression, we identify star and dog families in each year based on three-factor adjusted returns. T -statistics are reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 5: Three-factor stars

	1992	1993	1994	1995	1996	1997	1998
Constant	-3.39*** (-3.34)	-4.14** (-3.12)	-2.46*** (-2.48)	-3.64*** (-3.46)	0.07 (0.07)	-1.55* (-1.93)	-2.15*** (-2.99)
Cross-fund std. dev.	19.63 (0.35)	125.00** (2.20)	45.10 (0.82)	68.42 (1.28)	117.20** (2.08)	-66.62 (-1.21)	-4.92 (-0.14)
Family size	0.47** (2.05)	0.59** (2.31)	0.27 (1.27)	0.20 (0.99)	0.10 (0.50)	0.31* (1.71)	0.26 (1.74)
Number of funds	0.08 (0.16)	0.07 (0.15)	0.52 (1.44)	0.74** (2.22)	0.83** (2.47)	0.95*** (3.08)	1.07*** (4.04)
Turnover ratio	0.01 (0.04)	-0.39 (-0.78)	-0.67 (-1.24)	0.32* (1.66)	-1.02* (-1.90)	-0.67 (-1.43)	-0.30 (-0.93)
Expense ratio	59.75* (1.70)	8.53 (0.12)	-12.41 (-0.21)	-91.60 (-1.27)	-328.80*** (-4.23)	-4.67 (-0.09)	-33.38 (-0.64)
Front-end load	-1.46 (-0.15)	3.37 (0.32)	-13.43 (-1.18)	16.63 (1.64)	-8.89 (-0.72)	-22.56* (-1.88)	-13.86 (-1.30)
Lagged MS star dummy	2.08*** (4.25)	2.70*** (4.81)	2.69*** (4.89)	1.79*** (3.73)	1.99*** (3.41)	1.92*** (4.36)	0.66* (1.76)

The table examines which factors contribute to increasing the odds of producing MS stars. For each year from 1992 to 1998, we apply the following logit model to examine how family characteristics are related to the probability of producing star funds:

$$S_j = \beta_0 + \beta_1(\text{cross-fund std. dev.})_j + \beta_2(\text{family size})_j + \beta_3(\text{number of funds})_j + \beta_4(\text{turnover ratio})_j + \beta_5(\text{expense ratio})_j + \beta_6(\text{front-end load})_j + \beta_7(\text{lagged MS star dummy})_j.$$

Here, f is the index for fund family and S is an indicator variable that equals one if the family has at least one MS star fund and zero otherwise. *Cross-fund std. dev.* is the monthly average of the cross-fund standard deviations of three-factor adjusted returns for the family. *Family size* is the logarithm of monthly average of the relative family size (measured by the average TNA per member fund relative to that of the median family). *Number of funds* is the logarithm of monthly average of the total number of member funds managed by the family. *Turnover ratio*, *expense ratio*, and *front-end load* are the monthly averages of the corresponding statistics for the family. *Lagged MS star dummy* is a dummy variable that equals one if the family has at least one MS star fund in the previous year. In this regression, we identify MS star funds and MS star families in each year based on year-end Morningstar rankings. T -statistics are reported in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 6: Morningstar stars

6.5 Figure 1: future returns of star and nonstar family portfolios

Read:

- The average monthly four-factor adjusted return for all funds is negative, shown by the dashed line.
- HighStd-Large star portfolios perform poorly.
- The LowStd-Small star portfolio is the only star portfolio with clearly positive future performance.
- Nonstar portfolios generally do not display strong positive performance.

Economic message: Not all stars reveal skill. Stars produced by low-dispersion small families are more likely to be true stars.

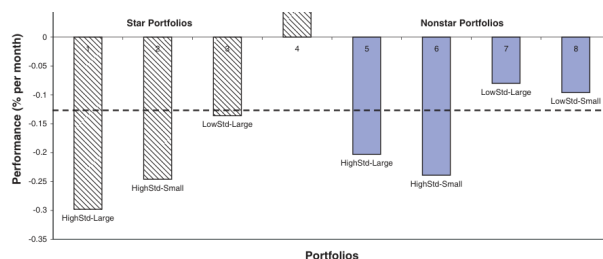


Figure 1
Average monthly four-factor adjusted returns for portfolios of star and nonstar families
The figure plots the average monthly four-factor adjusted returns for the eight portfolios reported in panel 1c of Table 9. The first four portfolios consist of funds belonging to star families and the last four portfolios consist of funds belonging to nonstar families. The dashed line represents the average monthly four-factor adjusted return (-0.125%) for the TNA-weighted diversified portfolio (including all funds in the sample).

6.6 Table 7 and Table 8: star-creating strategy and family performance

Read:

- Table 7 uses annual rebalancing; Table 8 uses monthly rebalancing.
- HighStd portfolios underperform LowStd portfolios, especially using four-factor adjusted returns.
- LowStd-Small-Star portfolios outperform other star portfolios in several comparisons.

- HighStd-Large-Star portfolios underperform LowStd-Star and some nonstar portfolios.
- The pattern is robust across one-factor, three-factor, and four-factor performance measures, though statistical significance varies.

Economic message: High-dispersion families may create stars but do not necessarily create value for investors. Future performance is strongest when stars arise in families with low ex ante star probability.

Portfolios	One-factor adjusted monthly returns (%)	Three-factor adjusted monthly returns (%)	Four-factor adjusted monthly returns (%)
Panel A: HighStd portfolios — LowStd portfolios			
All funds	-0.206 (-2.23)**	-0.076 (-1.16)	-0.159 (-2.63)***
Funds from star families	-0.175 (-1.24)	-0.012 (-0.12)	-0.183 (-2.13)**
Funds from non star families	-0.170 (-1.81)*	-0.106 (-1.68)*	-0.128 (-2.06)**
Panel B: Large portfolios — small portfolios			
All funds	-0.006 (-0.06)	-0.007 (-0.11)	-0.011 (-0.19)
Funds from star families	-0.088 (-0.62)	0.019 (0.18)	-0.036 (-0.41)
Funds from non-star families	0.027 (0.29)	-0.013 (-0.22)	0.015 (0.24)
Panel C: LowStd-small portfolio — other portfolios			
HighStd-Large	0.202 (1.90)*	0.089 (1.20)	0.169 (2.46)**
HighStd-Small	0.214 (1.70)*	0.105 (1.44)	0.160 (2.25)**
LowStd-Large	0.002 (0.02)	0.021 (0.36)	0.013 (0.23)
Panel D: HighStd-Large portfolio — other portfolios			
HighStd-Small	0.012 (0.09)	0.016 (0.19)	-0.010 (-0.13)
LowStd-Large	-0.200 (-2.18)**	-0.069 (-1.00)	-0.157 (-2.50)**
LowStd-Small	-0.202 (-1.90)*	-0.089 (-1.20)	-0.169 (-2.46)**

The table examines how cross-fund standard deviation and family size are related to family performance. At the beginning of each calendar year we form portfolios based on the two family attributes in the previous year: cross-fund standard deviation of three-factor adjusted returns and the number of member funds. The portfolios are held for exactly one year and rebalanced at the year-end. A family is defined as a high standard deviation (HighStd) family if its monthly average cross-fund standard deviation of three-factor adjusted returns in the previous year is above that of the median family. Otherwise the family is characterized as a low standard deviation (LowStd) family. Similarly a family is defined to be large (small) if its monthly average number of member funds in the previous year is above (below) that of the median family. We also form portfolios based on the above two factors separately for star families and nonstar families. For each portfolio we use the monthly average of TNA-weighted risk-adjusted returns during the period 1992 to 1998 to measure performance. In panel A, the performance of HighStd portfolios is compared to that of LowStd portfolios for all funds in the sample and separately for funds belonging to star and nonstar families. In panel B, the performance of large portfolios is compared to that of small portfolios for all funds in the sample and separately for funds belonging to star and nonstar families. Panels C and D compare the performance of LowStd-Small portfolio and HighStd-Large portfolio to that of other portfolios. To check for robustness, we use alternative return measures (one-factor adjusted returns, three-factor adjusted returns, and four-factor adjusted returns) to identify star families and to measure portfolio performance. The *t*-statistics are in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

High standard deviation families (HighStd-Large portfolios and HighStd-

Table 7: Annual rebalancing

Portfolios	One-factor adjusted monthly returns (%)	Three-factor adjusted monthly returns (%)	Four-factor adjusted monthly returns (%)
Panel A: HighStd portfolios — LowStd portfolios			
All funds	-0.197 (-2.12)**	-0.076 (-1.17)	-0.156 (-2.60)**
Funds from star families	-0.076 (-0.59)	-0.080 (-0.83)	-0.169 (-2.19)**
Funds from non-star Families	-0.223 (-2.38)**	-0.124 (-1.97)**	-0.160 (-2.52)**
Panel B: Large portfolios — small portfolios			
All funds	0.024 (0.27)	0.008 (0.13)	-0.009 (-0.15)
Funds from star families	-0.004 (-0.03)	0.019 (0.19)	-0.037 (-0.43)
Funds from non-star families	0.011 (0.12)	0.017 (0.29)	0.001 (0.02)
Panel C: LowStd-Small portfolio — other portfolios			
HighStd-Large	0.204 (1.96)*	0.091 (1.24)	0.174 (2.54)**
HighStd-Small	0.307 (2.37)**	0.140 (1.80)*	0.192 (2.55)**
LowStd-Large	0.021 (0.29)	0.023 (0.40)	0.023 (0.41)
Panel D: HighStd-Large portfolio — other portfolios			
HighStd-Small	0.103 (0.74)	0.048 (0.57)	0.018 (0.24)
LowStd-Large	-0.183 (-2.01)**	-0.068 (-1.03)	-0.150 (-2.47)**
LowStd-Small	-0.204 (-1.96)*	-0.091 (-1.24)	-0.174 (-2.54)**

The table examines how cross-fund standard deviation and family size are related to family performance. At the beginning of each month we form portfolios based on the two family attributes in the previous 12 months: cross-fund standard deviation of three-factor adjusted returns and the number of member funds. The portfolios are held for exactly one month and rebalanced at month-end. A family is defined as a high standard deviation (HighStd) family if its monthly average cross-fund standard deviation of three-factor adjusted returns in the previous 12 months is above that of the median family. Otherwise the family is characterized as a low standard deviation (LowStd) family. Similarly a family is defined to be large (small) if its monthly average number of member funds in the previous 12 months is above (below) that of the median family. We also form portfolios based on the above two factors separately for star families and nonstar families. For each portfolio we use the monthly average of TNA-weighted risk-adjusted returns during the period 1992 to 1998 to measure performance. In panel A, the performance of HighStd portfolios is compared to that of LowStd portfolios for all funds in the sample and separately for funds belonging to star and nonstar families. In panel B, the performance of large portfolios is compared to that of small portfolios for all funds in the sample and separately for funds belonging to star and nonstar families. Panels C and D compare the performance of LowStd-Small portfolio and HighStd-Large portfolio to that of other portfolios. To check for robustness we use alternative return measures (one-factor adjusted returns, three-factor adjusted returns, and four-factor adjusted returns) to identify star families and to measure portfolio performance. The *t*-statistics are in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 8: Monthly rebalancing

6.7 Table 9 and Table 10: true stars and performance skill

Read:

- Table 9 uses annual rebalancing and asks whether star portfolios outperform other portfolios.
- The diversified portfolio does not underperform the overall star portfolio, meaning naive star chasing is not profitable.
- LowStd-Small-Star portfolios outperform high-probability-star portfolios and many nonstar portfolios.
- Table 10 confirms similar patterns using monthly rebalancing, especially with four-factor adjusted returns.
- HighStd-Large-Star portfolios underperform LowStd-Small-Star portfolios in several comparisons.

Economic message: The star signal is useful only when interpreted together with the family's star-producing strategy. True skill is more likely when the star was unlikely to be produced mechanically.

diversified portfolio. The results indicate that the star portfolio (consisting

Table 9

Relative performance of fund family portfolios: annual rebalancing

Portfolios	One-factor adjusted monthly returns (%)	Three-factor adjusted monthly returns (%)	Four-factor adjusted monthly returns (%)
Panel A: Diversified portfolio — star and nonstar portfolios			
Star portfolio	0.050 (0.47)	0.018 (0.23)	0.084 (1.23)
Non-star portfolio	0.029 (0.38)	0.026 (0.43)	0.001 (0.01)
Panel B: LowStd-Small-star portfolio — other portfolios			
Star portfolios:			
HighStd-Large	0.431 (2.65)***	0.361 (2.27)**	0.378 (3.64)***
HighStd-Small	0.344 (1.86)*	0.354 (2.06)**	0.326 (2.78)***
LowStd-Large	0.264 (1.71)*	0.351 (2.25)**	0.216 (2.14)**
Nonstar portfolios:			
HighStd-Large	0.384 (2.68)***	0.411 (2.71)***	0.283 (2.97)***
HighStd-Small	0.460 (2.68)***	0.421 (2.72)***	0.319 (3.27)***
LowStd-Large	0.229 (1.88)*	0.312 (2.17)**	0.160 (1.91)*
LowStd-Small	0.249 (1.87)*	0.300 (2.03)**	0.176 (1.94)*
Panel C: HighStd-Large-Star portfolio — other portfolios			
Star portfolios:			
HighStd-Small	-0.087 (-0.50)	-0.007 (-0.06)	-0.052 (-0.49)
LowStd-Large	-0.167 (-1.13)	-0.010 (-0.10)	-0.162 (-1.78)*
LowStd-Small	-0.431 (-2.65)***	-0.361 (-2.27)**	-0.378 (-3.64)***
Nonstar portfolios:			
HighStd-Large	-0.047 (-0.35)	0.049 (0.53)	-0.095 (-1.10)
HighStd-Small	0.029 (0.18)	0.060 (0.62)	-0.059 (-0.66)
LowStd-Large	-0.202 (-1.71)*	-0.049 (-0.60)	-0.218 (-2.88)***
LowStd-Small	-0.182 (-1.41)	-0.062 (-0.71)	-0.202 (-2.48)**

The table documents the relative performance between various portfolios. At the beginning of each calendar year we form portfolios based on the following family attributes in the previous year: cross-fund standard deviation of three-factor adjusted returns, the number of member funds, and the star identify. The portfolios are held for exactly one year and rebalanced at year-end. A family is defined as a high standard deviation (HighStd) family if its monthly average cross-fund standard deviation of three-factor adjusted returns in the previous year is above that of the median family. Otherwise the family is characterized as a low standard deviation (LowStd) family. Similarly a family is defined to be large (small) if its monthly average number of member funds in the previous year is above (below) that of the median family. A family is defined as a star family if it has at least one star fund under management based on the monthly average of risk-adjusted returns in the previous year. We denote all portfolios by the family characteristics defined above. LowStd-Small-Star refers to the portfolio consisting of all funds belonging to star families with high standard deviation and small size. Similar notation applies to other portfolios. For comparison purposes, we also construct a diversified portfolio that includes all funds in the sample. For each portfolio we use the monthly average of TNA-weighted risk-adjusted returns during the period 1992 to 1998 to measure performance. Panel A reports the performance difference between the diversified portfolio and the star (non-star) portfolio. In panel B we compare the performance difference between the LowStd-Small-Star portfolio and other portfolios. In panel C we compare the performance difference between the HighStd-Large-Star portfolio and other portfolios. To check for robustness, we use alternative return measures (one-factor adjusted returns, three-factor adjusted returns, and four-factor adjusted returns) to identify star families and to measure portfolio performance. The t -statistics are in parentheses.

Table 9: True stars, annual rebalancing

Portfolios	One-factor adjusted monthly returns (%)	Three-factor adjusted monthly returns (%)	Four-factor adjusted monthly returns (%)
Panel A: Diversified portfolio — star and nonstar portfolios			
Star portfolio	0.012 (0.13)	-0.014 (-0.19)	0.017 (0.27)
Nonstar portfolio	0.047 (0.60)	0.028 (0.47)	0.017 (0.29)
Panel B: LowStd-Small-Star portfolio — other portfolios			
Star portfolios:			
HighStd-Large	0.184 (1.21)	0.252 (1.47)	0.230 (2.24)**
HighStd-Small	0.321 (1.62)	0.236 (1.31)	0.226 (1.86)*
LowStd-Large	0.140 (1.06)	0.231 (1.37)	0.089 (0.88)
Nonstar portfolios:			
HighStd-Large	0.341 (2.53)**	0.344 (2.13)**	0.275 (2.74)***
HighStd-Small	0.384 (2.36)**	0.369 (2.22)**	0.272 (2.55)**
LowStd-Large	0.123 (1.08)	0.223 (1.44)	0.115 (1.29)
LowStd-Small	0.125 (0.98)	0.222 (1.40)	0.119 (1.22)
Panel C: HighStd-Large-Star portfolio — other portfolios			
Star Portfolios:			
HighStd-Small	0.136 (0.71)	-0.015 (-0.13)	-0.004 (-0.04)
LowStd-Large	-0.044 (-0.34)	-0.020 (-0.20)	-0.142 (-1.75)*
LowStd-Small	-0.184 (-1.21)	-0.252 (-1.47)	-0.230 (-2.24)**
Non-star Portfolios:			
HighStd-Large	0.156 (1.17)	0.093 (1.01)	0.045 (0.55)
HighStd-Small	0.200 (1.25)	0.118 (1.19)	0.042 (0.48)
LowStd-Large	-0.062 (-0.54)	-0.028 (-0.35)	-0.116 (-1.70)*
LowStd-Small	-0.059 (-0.47)	-0.029 (-0.33)	-0.111 (-1.43)

The table documents the relative performance between various portfolios. At the beginning of each month we form portfolios based on the following family characteristics in the previous 12 months: cross-fund standard deviation of three-factor adjusted returns, the number of member funds, and the star identify. The portfolios are held for exactly one month and rebalanced at month-end. A family is defined as a high standard deviation (HighStd) family if its monthly average cross-fund standard deviation of three-factor adjusted returns in the previous 12 months is above that of the median family. Otherwise the family is characterized as a low standard deviation (LowStd) family. Similarly a family is defined to be large (small) if its monthly average number of member funds in the previous 12 months is above (below) that of the median family. A family is defined as a star family if it has at least one star fund under management based on the monthly average of risk-adjusted returns in the previous 12 months. We denote all portfolios by the family characteristics defined above. LowStd-Small-Star refers to the portfolio consisting of all funds belonging to star families with high standard deviation and small size. Similar notation applies to other portfolios. For comparison purposes, we also construct a diversified portfolio that includes all funds in the sample. For each portfolio we use the monthly average of TNA-weighted risk-adjusted returns during the period 1992 to 1998 to measure performance. Panel A reports the performance difference between the diversified portfolio and the star (nonstar) portfolio. In panel B we compare the performance difference between the LowStd-Small-Star portfolio and other portfolios. In panel C we compare the performance difference between the HighStd-Large-Star portfolio and other portfolios. To check for robustness we use alternative return measures (one-factor adjusted returns, three-factor adjusted returns, and four-factor adjusted returns) to identify star families and to measure portfolio performance difference. The t -statistics are in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Table 10: True stars, monthly rebalancing

7 Short summary

- The paper studies how mutual fund family structure affects flows, strategies, and performance.
- A star fund increases inflows to the star fund and to other funds in the family.
- The family-level spillover makes a star more valuable than in a stand-alone fund.
- Families with high cross-fund return dispersion and more member funds are more likely to produce stars.
- This suggests that families can pursue star-creating strategies.
- Lower-ability families may rationally pursue such strategies because investor flows respond convexly to extreme winners.
- Naively investing in star families does not necessarily generate superior future returns.
- Stars are more informative when they arise in low-dispersion small families, where the ex ante probability of producing a star is low.
- The main lesson is that investor flows, family strategy, and fund performance should be analyzed at the family level as well as at the fund level.

8 Conclusion

8.1 Core conclusion

Nanda, Wang, and Zheng show that mutual fund families create economically important spillovers. A star fund attracts money not only to itself but also to sibling funds. This makes the family-level payoff to star

creation much larger than the payoff to a stand-alone star fund.

8.2 Economic intuition

Investors search imperfectly and pay attention to highly visible winners. A star fund can attract investor attention to the entire fund family. Since flows respond convexly to strong performance, families may benefit from increasing the probability of producing at least one star. This can be done by offering more funds or by increasing dispersion in strategies across member funds.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to connect three mechanisms that were usually studied separately: convex flow-performance sensitivity, family-level spillovers, and strategic variation in investment styles. It shows that the family structure can change managerial incentives and that not every observed star is equally informative about skill.

8.4 Investor takeaway

Investors should not simply chase funds from star families. A star from a high-dispersion large family may be the outcome of a lottery-like strategy. A star from a low-dispersion small family is more likely to indicate genuine investment ability.